

Ensemble CAD System for Acute Lymphoblastic Leukaemia Classification

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Abstract—Acute Lymphocytic leukaemia is a type of blood cancer which affects white blood cells (WBC) and can cause damage to bone marrow. This disease is diagnosed with a complete blood count (CBC) and a bone marrow aspiration, and it can be lethal if not caught early. In general, manual procedures such as microscopic inspection of stained sample slides are employed to diagnose leukemia. However, manual methods are time-consuming, less accurate, and might result in errors due to numerous human variables such as stress and exhaustion. So, this study discusses the methodology of computer-aided diagnosis systems, including enhancement, augmentation, ensemble, feature extraction, and the usage of CNN-based methods. In proposed method, two classes (Healthy/infected) are discretized using the ALL- IDB-2 dataset, which provided segmented microscopic images. Classification models like VGG16, VGG19, XCEPTION, and Mobile-Net are used with an accuracy of 96 % for vgg16, 94 % for VGG19, 92 % for XCEPTION, and 96 % for Mobile-Net. In this study, Mobile-Net performed better than the other three models. Finally, all the models are utilized by using ensemble method with highest accuracy of 98%.

Keywords— *Acute Lymphoblastic leukemia, Ensemble classifier, CNN.*

I. INTRODUCTION

Acute lymphocytic leukaemia (ALL) is a kind of blood cancer which affects white blood cells. A bone marrow disease occurs when abnormal white blood cells begin reproducing in the bone marrow. In the battle against sickness, these cells don't work well. It builds up in the bone marrow, preventing the production of other normal blood cells, resulting in anaemia, bleeding, and recurring infections. As the aberrant cells divide, they move into the bloodstream, occasionally forming malignant tumours and causing damage to organs including the kidneys and liver. Acute leukaemia is a quickly progressing disease that mostly affects developing cells (not fully developed). ALL is more common in children, whereas AML is more common in adults, but it can also affect children and adolescents [1].

Early identification of a form of leukaemia is critical in determining the best treatment options for that type. A complete blood count is used to diagnose it (CBC). If the count is abnormal, the patient will have a bone marrow biopsy. As a result, morphological bone marrow and peripheral blood slide analysis are used to validate the existence of leukemic cells. A pathologist will examine some cells under light in order to distinguish aberrant cells from their specific types and subtypes of leukaemia. Microscope is used to look for abnormalities in the nucleus or cytoplasm of cells. This is critical in clinical prediction of disease behaviour and prognosis in order to choose the best treatment for the patient [1].

The French American-British (FAB) system classifies acute leukemia into two types: acute lymphoblastic leukemia (ALL) and acute myelogenous leukaemia (AML). The following is the FAB classification.

- **L1** - The cells are uniformly tiny and small. The nuclei are spherical, and small and invisible fractures nucleoli are prevalent. The vacuoles in cytoplasm are frequently tiny and non-existent [2].
- **L2** - This category contains a variety of cell types. The nuclei are extremely rare and frequently fractured. There are usually one or more main nucleoli present. Despite its adaptability, the cytoplasmic volume is frequently overrun and may contain vacuoles. [2].
- **L3** -The cells are medium-sized and uniform in size. These are the nuclei that are irregularly shaped and ovaly. The cytoplasm's capacity is central, and substantial vacuoles are present [2].

This procedure, however, is defective, and it is a difficult assignment for the haematologist to perform. As a result, a haematologist requires a computer-aided diagnosis system (CAD) capable of overcoming previous system constraints and distinguish immature blood cells from healthy mature

cells. Deep learning, on the other hand, can read and extract high-quality features separation simultaneously. As a result, we propose a new Convolutional Neural Network structure to distinguish between normal and abnormal blood cell pictures the authors introduced two alternative techniques to using power CNNs in recent studies. The first is the most typical strategy, which involves training with a significant amount of data. The second method is to use pre-trained sets to transfer learning.

The standard approach for acquiring lymphoblasts in microscopically coloured images can be broken down into the components below [2]. Cells are segregated by their backgrounds utilizing algorithms based on cell characteristics (such as shape, colour, and interior momentum). Identification of white cells is important. The two types of cells are white cells and red cells. Color information can be used to look for the existence of the nucleus via the classifier [2].

Lymphocytes are distinguishable from other types of white blood cells by examining the shape of a nuclear nucleus (e.g., a dark-colored nuclear nucleus that may be eccentric in some ways, cytoplasm in modest amounts) [2]. Identification of candidate lymphoblasts is done by using morphological modification study of lymphocyte that can reveal selected lymphoblasts in a group of lymphocytes [2].

The main aim of the current research is to analyse the different AI techniques for the detection of lymphoblast cells and find a more accurate method for this. The paper is divided as follows: the second section contains related work done by other researchers in this field. The next section is about methods and various classification techniques used for classifying the WBCs. In the next section, the result of the research is presented. Finally, a brief discussions and comparisons of few previous research papers are highlighted.

II. LITERATURE REVIEW

Several methods already being used over the years to enhance the efficiency and accuracy in different types of parameters. Most of the authors in their literature have used techniques of machine learning including the K-means clustering algorithm basically for the detection and for the classification of images of the blood cell [2-3]. In almost all the cases, common mathematical characteristics including strength, entropy, comparisons, correlations, has taken out and presented as an input into the models of machine learning. It can be seen that the traditional machine learning methods have its own drawback like time-consuming development and, in particular, the requirement to determine what type of features should be used to increase the accuracy [3]. The most used techniques for the classification are Naive Bayesian (NB), Support Vector Machine (SVM), Random Forest algorithm (RF), K-Nearest Neighbors (KNN), Multilayer Perceptron (MLP) and Radial Basis Function Network (RBFN) [4].

Mohaprata et al. [5] proposed a method having the shape characteristics i.e., Hausdorff Dimension and contour signature, basically used for the classification of the lymphocytic cells and for the classification SVM was used which gave a 93 percent level of precision.

Mohapatra et al. [6] suggested a method to improve the accuracy of the ALL diagnostic by analyzing morphological and textural features with the help of ensemble classifier and it was observed that the ensemble classifier leads to 99% accuracy in comparison to other classifiers i.e. KNN, NB, MLP, RBFN, SVM.

Patel and Mishra et al. [7] proposed a method in which an automated approach for the leukemia detection is used. K-means clustering was used for detection of white blood cells, and SVM was used for classification, with an accuracy of 93.57 %.

Bhattacharjee et al. [8] proposed a method in which the identification of acute lymphoblastic is done and watershed transforms is used for the segmentation and then comes the morphological operation. Morphological features are extracted, and for classification, GMM and binary search tree are used with 95.56 % of accuracy.

Vincent et al. [9] proposed a method in which a novel approach is used for acute leukemia type classification using the classifier named as sequential neural network. In this, basically two steps neural network classifiers is applied i.e. First Neural Network Classifiers and Second Neural Network Classifiers. The first classification process performs admirably in distinguishing between normal and abnormal cells. As a result, the SNN's accuracy reached 97.7 %.

Singhal et al. [10] proposed a method for detecting ALL using descriptors of texture derived from the nucleus image. They essentially used two texture features called Local Binary Pattern (LBP) and Gray Level Co-occurrence Matrix (GLCM) for automatic detection of ALL, with LBP features accuracy of 93.8 % and with GLCM accuracy of 87.30 % is obtained.

Rawat et al. [11] have proposed a technique to differentiate between the ALL affected cells from the Healthy cells. This technique firstly extracted the leukocytes which are separated from the other blood cells, and then the lymphocytes are extracted. In this scenario a CAD system is used for the detection of the leukemia based on GLCM and shape features and classified by the SVM with 89.8% accuracy.

Rehman et al. [12] proposed a method in which robust and deep learning technique along with the CNN were used to train the model to achieve the accurate classification result and the classifiers used were Naive Bayesian, KNN, SVM and achieved accuracy is 97.78%.

Shafique et al. [13] proposed a method to demonstrate an approach to segmentation using watershed transform and two simpler classification models to diagnose the presence of ALL precisely in a blood smear.

Acharya et al. [14] proposed a method to distinguish between a typical blood as well as an abnormal blood sample. For classification different classifiers are compared with the highest accuracy of 98.6%.

III. METHODOLOGY

TABLE I. COMPARISON OF THE PROPOSED WORK WITH THE DIFFERENT WORKS BY OTHER AUTHORS FOR THE CLASSIFICATION OF LYMPHOCYTIC IMAGES

S. No.	Ref. No. / Year of Publication	Extracted Features	Datasets	Classifiers	Accuracy (%)
1	[5], 2011	Texture, Shape and Color features	ALL-IDB	Support Vector Machines	93
2	[6], 2014	Texture and Shape features	ALL-IDB	Ensemble (Radial Basis Functions Neural Network + Support Vector Machines + Multilayer Perceptron Neural Network + K-nearest neighbor + Naïve Bayes)	99
3	[7], 2015	Texture, Shape Color and Statistical features	ALL-IDB	Support Vector Machines	93.75
4	[8], 2015	Morphological features	ALL-IDB1	GMM and Binary search tree	95.56
5	[9], 2015	Texture features	ALL-IDB2	MLP	97.70
6	[10], 2016	Texture features	ALL-IDB1	Support Vector Machines	93.8
7	[11], 2017	Shape based features	ALL-IDB1	Support Vector Machines	89.8
8	[12], 2018	CNN features	ALL-IDB1	CNN	97.78
9	[13], 2018	Color features	ALL-IDB2	Deep Neural Network	99.50
10	[14], 2019	Shape, visual and texture features	ALL-IDB1	Decision tree, Random Forest, Naïve Bayes and K-nearest neighbors	98.6
11	[15], 2021	CNN based features	ALL-IDB	CNN	99.33
12	[16], 2015	Texture, Shape and Color features	ALL-IDB2	Dempster-Shafer (Support Vector Machines + Multilayer Perceptron Neural Network)	96.72
13	[17], 2014	Texture, Shape and Color features	ALL-IDB1	Support Vector Machines	93.63
14	[18], 2012	Texture, Shape and Color features	ALL-IDB	Support Vector Machines	93.50
15	[19], 2018	Texture, Shape and Color features	ALL-IDB1	Support Vector Machines	>94.00
16	[20], 2017	Texture and Shape features		MLP	97.20
17	[21], 2018	CNN based features	ALL-IDB	CNN	96.6
18	[22], 2019	SVMs applying handcrafted feature	ALL-IDB	CNN	81.74
19	[23], 2014	Color and Statistical Features	ALL-IDB	Support Vector Machines	93
20	Proposed Method	CNN based features	ALL-IDB2	Ensemble (VGG16 + VGG19 + Mobile Net + Xception)	98

B. Data Augmentation

Data augmentation is a methodology, through which one can increase the amount of data by adding some moderate copies of already existing data by rotating, zooming, shearing, etc. or creating some new artificial data from the existing dataset. Data augmentation works as a regularizer and helps in avoiding the case of over fitting of model, while a model is trained with small amount of dataset. As the dataset used contains only 260 images, therefore augmented training data has been used to avoid over fitting problem [25].

C. Convolution Neural Network

A convolutional neural network in deep learning is an algorithm designed to classify or differentiate images from one other by assigning biases and learnable weight. The stage of pre-processing needed in a Convolution Network is not as much as compared to other classification algorithms. The convolution network possesses the ability to learn the filters

A. Database Description

In the proposed work, the famous ALL-IDB dataset is used which is divided into: ALL-IDB1 and ALL-IDB2. ALL-IDB2 is utilized, where Lymphoblast accounting for 50% of the total. There are 260 images in this dataset. There are 130 images which are ALL affected and 130 pictures which are NOT ALL affected. The picture dimensions are the sole change between ALL-IDB1 and ALL-IDB2. Some of the sample images from the dataset are shown in Fig.1 [24].

or characteristics, whereas in primitive methods the filters are trained. A convolution neural network is made up of input layer, followed by hidden layer and at last the output layer. The hidden layer in the network comprises of convolution layers, fully connected layers, pooling layers and normalization layers. The output of hidden layer works as an input to activation function which decides whether neuron should be activated. Soft-Max layer has been used as an output layer activation function, to classify the images [25].

D. Classification Models

In recent times, deep learning has been contributing in a number of fields like medical image analysis, agriculture, etc. for various purposes. In order to make an efficient and reliable model have been preferred to use pretrained models like VGG16, VGG19, Mobile Net and Xception in the project, rather than starting from scratch. At last, the ensembling of all these models was done to get a better efficient model. The use of convolution network has been

done to extract features from the images as well as to classify the images into classes of ‘ALL Affected’ and ‘Healthy’, using soft-max activation function in dense layer which will work as an output layer.

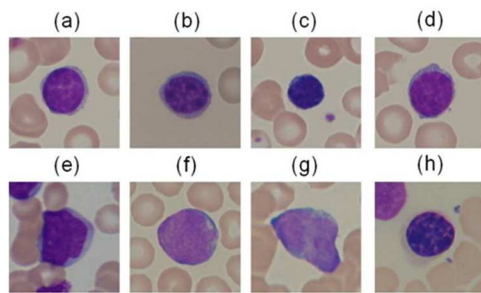


Figure 1. Some of the sample images from ALL-IDB2. (a-d) Non ALL patients/ healthy cell images, (e-h) ALL infected cell images.

- *VGG16*

VGG16 is a convolution neural network designed for ILSVRC competition. The VGG16 is amongst one of the best performing vision model. The VGG16 consist of 16 layers out of which there are thirteen convolution layers, five max pooling layers and three fully connected layers. The model is designed to have a convolution network of 3x3 filter with a stride 1 and padding of 1 which is followed by maxpool layer with 2x2 filter of stride 2. This kind of ordering of convolution layers and maxpooling layers is observed to be followed uniformly all through the architecture [26].

- *VGG19*

VGG19 is a CNN architecture designed to win the ILSVRC imageNet competition. The VGG19 takes an image input size of 224×224. The Network has learned, from a large number of ImageNet dataset with representations of rich features for a broad range of images. It is 19 layers deep and contains sixteen layers of convolution, three layers of fully connected, five layers of max pool and one layer of soft-max [27].

- *MobileNet*

MobileNet is a 28 layers deep CNN architecture that is designed to be used by embedded vision applications and mobile. MobileNet is made-up of streamlined architecture, it uses depth wise separable convolutions whose main aim is to build lightweight neural networks possessing low level latency for embedded device or mobile [28].

- *Xception*

Xception is 71 layers deep convolutional neural network and it is called as extreme inception, as it takes inception to extreme. It involves depth wise separable convolutions, at first on each depth map filters are applied and finally compressing the input space with help of 1x1 convolution which is applied over the depth of network [29].

- *Ensemble Classifier*

An ensemble model is the combination of multiple learning algorithms which are combined to obtain their collective performance. The ensembling of models improves the performance of existing models. The main aim is to get a reliable model that can use the features of all other models. In most of applications, deep learning models individually proved to be competent but ensembling provides scope to use a group of deep learning models to perform the same task with better performance [30].

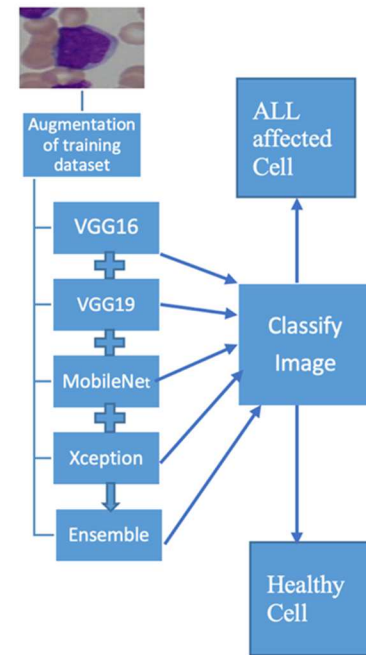


Figure 2. Architecture of Proposed classification model.

IV. RESULTS

A. Performance of proposed method

This section discusses the results of the experiments conducted during the research by implementing different classification models as VGG-16, VGG-19, XCEPTION, Mobile Net and ensemble model.

- The VGG-16 model's performance: In this study, a pre-trained VGG-16 model is retrained on the leukocyte images of ALL-IDB2 dataset. This is done by using augmented training and validation sets in order to get the desired result from proposed model. In Experiment 1, 96 % accuracy is obtained by using VGG-16. The performance of VGG-16 can be shown by its confusion matrix as given in Fig. 3.
- The VGG-19 model's performance: To meet the goal of the VGG-19 model, a pre-trained VGG19 model was retrained on ALL-IDB 2 dataset using the expanded

training and validation sets in this trial. In Experiment 2, 94 % of accuracy is achieved using VGG-19 model.

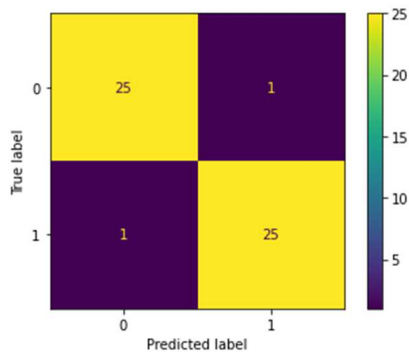


Figure 3. Confusion Matrix for leukaemia classification using VGG-16.

In Fig. 4, the performance of VGG-19 is shown with the detail of number of correctly classified data.

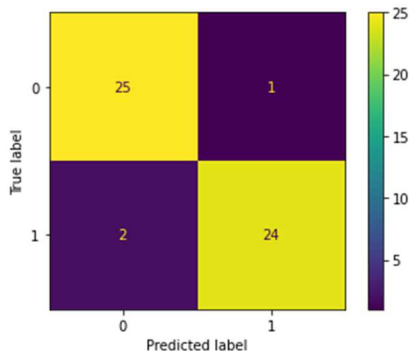


Figure 4. Confusion Matrix for leukaemia classification using VGG-19.

- Xception model performance: To fulfil the goal of the XCEPTION model, a pre-trained XCEPTION model was retrained on ALL-IDB 2 dataset images using the expanded training and validation sets in this trial. In Experiment 3, 92 % accuracy is obtained using XCEPTION as shown in Fig. 5.

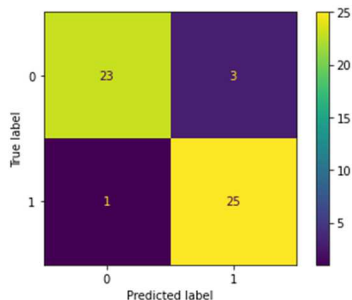


Figure 5. Confusion Matrix for leukaemia classification using XCEPTION.

- MobileNet model’s performance: Same set of images are applied to MobileNet for enhancing the classification

accuracy. In Experiment 4, 96 % accuracy achieved as depicted in Fig. 6.

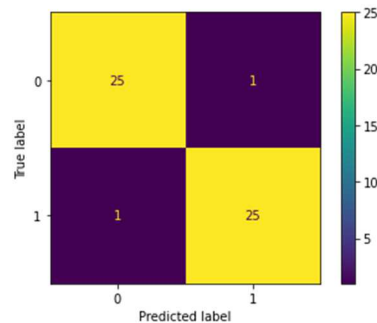


Figure 6. Confusion Matrix for leukaemia classification using MobileNet.

- Ensemble Classifier performance: In the Experiment 5, the combination all of these models was done into a weighted average ensemble model, which tends to improve the classification accuracy to 98 % as shown Fig. 7.

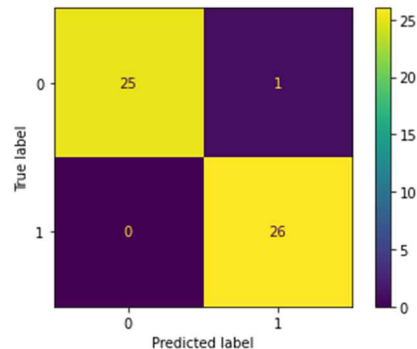


Figure 7. Confusion Matrix for leukaemia classification using Ensemble classifier.

The performance of ensemble method can be seen in the ROC curve graph for all the aforementioned models in Fig. 8.

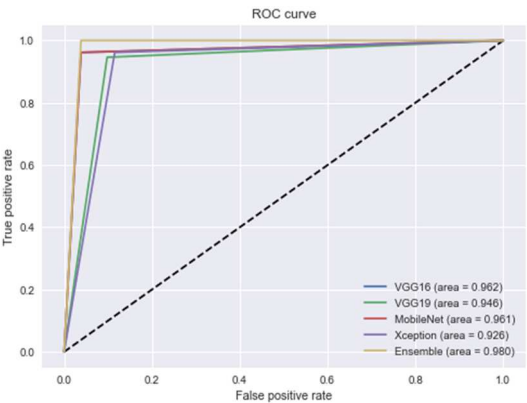


Figure 8. ROC curve representing all models.

V. CONCLUSION

In this study, a quick overview is provided for leukaemia classification and emphasized several AI based techniques presented by various researchers. Acute lymphoblastic leukaemia is a disease that is fatal that must be remedied as soon as possible. In present work, various strategies for early detection are examined. Different pretrained CNN models are used to extract the feature of images and classify them into two classes of healthy or infected. At last, an ensemble model with weighted average is used which performed better than other models. In future, it is proposed to use other techniques like object detection methods on different dataset and also to use other deep learning techniques to increase the reliability of model.

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